

SEQUIN: A Network Science and Physics-based Approach to Identify Sequential N-k Attacks in Electric Power Grids

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Abstract

The electric grid is a vital infrastructure that supplies power on which modern society depends, and maintaining its reliable service and resilient operation is essential. The grid's performance typically relies on a few key components. However, efficiently finding these components is challenging, due to the geo-distributed scale of the grid, complex physics governing power flows, and automated network response. Realistically identifying these key components must also consider the temporal aspect of how failures affect the network. In this paper, we address the problem of identifying worst-case disruptions to the grid, under the sequential failure of components. We present SEQUIN, a framework leveraging network science principles and physics-based constraint optimization to explore such failures in the grid. We formulate the problem using a sequential N-k interdiction model, which provides a methodology to explore and capture interactions between the failures and network response. Our approach defines several network properties to assess the contribution of each component towards its operation, and provides an efficient guided exploration of attacks. We also provide a toolkit to help reason about the impact on the grid. Extensive experiments on multiple benchmark grid networks are conducted to show the efficacy of our approach and demonstrate how the varying the sequence of attacks can result in different levels of disruption.

CCS Concepts

• **Hardware** → **Smart grid**; • **Computing methodologies** → *Network science*; • **Networks** → *Network dynamics*.

Keywords

resilience, power grid, N-k interdiction, sequential failure

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1 Introduction

The electric grid is a critical infrastructure system responsible for generating, transmitting, and distributing power to our cities and communities. It consists of an interconnected system of generators which serve as sources of electricity (e.g., power plants, wind turbines); buses, transmission lines, and distribution lines which form a transport network to carry electricity; substations which help manage the flow of electricity for distribution; and end users who consume power for their daily needs (e.g., homes, restaurants, etc.). The grid also contains systems for real-time monitoring and control such as Supervisory Control and Data Acquisition (SCADA) to help ensure power stability and balance. However, recent events such as Hurricane Harvey in 2017 [41] and the Moore County substation terror attack in 2022 [38] have revealed the fragility of the electric grid, and how its failure from natural disaster or man-made attacks can lead to disruptions with major consequences.

Many grid systems are dependent on a few key components (generators and lines) to effectively serve end users [6, 11], but identifying such elements efficiently (e.g., for ruggedization), in particular combinations, is challenging. One factor is the geo-distributed scale and complexity of the network, which cover large areas with thousands of nodes and links. The physics governing power flows and their interdependencies in the grid also contribute to the difficulty in predicting the impact of a failing generator or line: non-essential components for normal operation may become crucial as more failures occur. Moreover, the automated response of grid networks based on physics and engineering may potentially lead to undesirable power flows (e.g., blackout) that can occur before grid operators can intervene. Within the literature, it often assumed that these sets of component fail simultaneously (or near simultaneously). However, we observe that considering the time between failures is also essential in analyzing grid performance, i.e., understanding *when* failures would occur is just as important as *where* they occur.

Despite these challenges, we can leverage multiple structural and behavioral aspects of the infrastructure to provide insights into the vulnerabilities of the grid. First, the network topology and physical characteristics of its components (e.g., capacities of lines, locations of load demands) can be used to find bottlenecks in the grid. We can also study the network's response to failures over time to capture its behavior and the contribution of each component in supporting network operation. In doing so, we can discover the spatiotemporal dependencies between different components, and

the conditions under which they become more important. Collectively, these provide an intuition for how failures in the grid can affect its ability to deliver power to end users.

This paper presents SEQUIN, a network science and optimization based approach to examine how failures in the grid infrastructure can lead to large disruptions in service. We explore this problem through the framing of a *sequential N-k interdiction model*, which conceptualizes two opposing players whose goals are to produce or protect against worst-case disruptions. In this context, we look to develop methods that relax the assumption of simultaneous failures of components, and show how key components of the grid can be efficiently identified based on their impact in the network. We define several network properties and metrics that can be monitored to characterize the current state of the grid, and its capacity to serve power. Physics-based constraint optimization models are also leveraged to capture subsequent network behavior, and predict impacts of future failures. Our key contributions include:

- The SEQUIN framework utilizing network science and bi-level constraint optimization to identify sequential attacks in the grid
- A mathematical formulation of the sequential N-k interdiction problem as a multi-level, physics-based optimization model
- A detailed validation of the SEQUIN approach using multiple academic benchmark grid networks
- A comparison of sequential failures with simultaneously failures which demonstrate that the sequence of an attack matters.

We organize the rest of the paper as follows: Sec. 2 examines methodologies in analyzing the resilience of the power grid, and identifying crucial components of networks. Then, in Sec. 3 we introduce key concepts in the electric grid domain and formalize the sequential network interdiction problem. The SEQUIN solution approach is then detailed in Sec. 4. Our experimental validation is presented in Sec. 5, and we discuss future directions in Sec. 6.

2 Analyzing Resilience of the Grid

We next discuss the problem of identifying the key components in the operation of the grid infrastructure. To this end, we present background for N-k interdiction, and methodologies used to study worst-case disruptions to the grid and their limitations.

2.1 Background and Challenges

The problem of identifying the components whose failure leads to the worst-case disruption in a network is considered NP-hard [3]. Hence, some research efforts have focused towards techniques utilizing network science and machine learning to gain insights. Typical topological properties, which focus primarily on the structure of nodes and edges in the network, include power law relationships with degree [20, 30], and betweenness [1]. Empirical properties, which are normally derived by running simulations to study impact, include centrality [5], percolation [12, 16], and criticality [2, 14, 15]. Indices for resilience have also been proposed to help quantify measure the reliability and robustness of different events [22, 27, 32]. In addition to these traditional methods of studying networks, work in applying machine learning to characterize and identify vulnerabilities have helped to extract more complex relationships between network structures (nodes/edges) and behaviors (phenomena of

interest). Techniques such as Bayesian networks [19, 44], spectral clustering [29], and support vector machines [13] have been adopted in this manner. Emerging deep learning architectures have also been used to study the complex physics driving network behavior, from physics-informed neural networks [33, 42] to deep reinforcement learning [9, 18], and more.

The N-k Network Interdiction Model. A common model used to quantify the resilience of a network to failure is the *N-k interdiction model*, which is based on the zero-sum Stackelberg game [25, 34, 36, 45]. In this context, the goal is to find the k components of a network of size N whose simultaneous (or near-simultaneous) failure most impacts the grid. A typical setting considers two opposing players: an *attacker* who aims to select k components to disable to cause the most disruption in service; and a *defender* who aims to minimize the degradation of service caused by the attacker's actions (e.g., by rerouting power flow, adjusting generator output). The degree of service disruption is normally measured using *load shed*, which represents the amount of power that needs to be cut from the network to maintain system operation.

For the attacker, one key factor is the extent of knowledge about the system. Models assuming perfect knowledge, where the attacker is fully aware of the system's state and response, are often explored under worst-case conditions [24, 31, 34]. Work in studying attackers with imperfect knowledge, e.g., due to dynamic/unpredictable demand patterns or limited access to system topology and data have also been considered [23, 40]. Attackers can also be restricted on their ability to interdict specific components in the network. Efforts such as [34, 35] extend the traditional interdiction model with spatial constraints on the attacker to better model credible attacks, while others like [8, 26] examine ruggedized infrastructures with protected generators/lines. On the other hand, the defender's power is based on their ability to respond to attacks and maintain system functionality. This can be based on a defender who has full knowledge of the network by default [31, 34], or one that operates in uncertainty, e.g., due to false data [37, 43] or unidentifiable attacks [28]. The adaptability of the system, such as modifying topologies through line switching [10, 46] has also been explored.

Key Limitations. A common assumption made in many interdiction models is that the failure of its key components is simultaneous. We argue, however, that considering *sequential* failures instead, in which components are disabled one after another, offer a more realistic depiction of the attacks on the grid and their subsequent response. This comes from the observation that attackers are unlikely to be able to fail their selected components at the same time before an automated network response; other natural events, e.g., power trips, also follow this pattern. Moreover, such sequential failures have a *potential* in some scenarios to cause higher worst-case disruptions than their simultaneous counterparts, as we will demonstrate later; this is due to the additional degree of freedom in selecting *when* failures occur as well as *where*. As a result, new analyses to study and understand the impact of sequential failures in the grid is also required.

2.2 Overview of the SEQUIN Approach

To comprehensively study sequential failures and the impact in the grid, an attacker would need to explore all possible sequences

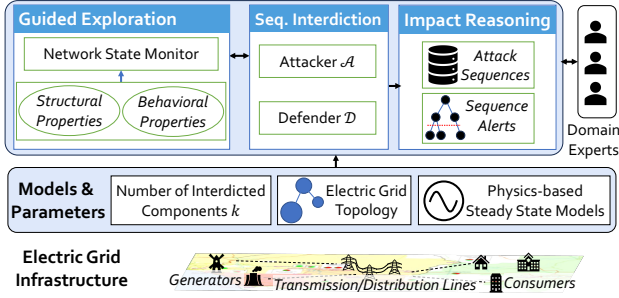


Figure 1: Overview of the SEQUIN Architecture

of failures. However, doing so quickly becomes computationally intractable as the scale of the network and size of an attack grows. Thus, our goal is to develop scalable and effective strategies to explore attacks on the grid. To this end, we present SEQUIN: a network science and optimization based approach to generate sequential attacks and reason about their impact, using the framework of a SEQUENTIAL INTERDICTION model. The intuition behind SEQUIN lies in how structural and behavioral properties derived from network science principles can help quantify the stability and adaptability of the grid when failures occur. Efficient physics-based models for power flow are employed to provide a realistic approximation of the evolving dynamics¹ in the network. This provides insight into the role that network components play in the operation of the system.

Fig. 1 shows the general workflow of SEQUIN. Let k , the total number of components that an attacker can choose to fail, be a fixed parameter. We assume that the underlying grid network’s topology and physical characteristics of its components are provided a priori, e.g., from public sources like [39], or by grid operators. Our approach leverages three core modules to address the problem of identifying worst-case sequential failures in the grid. First, the **Guided Exploration** module searches for potential attacks in the grid based on a number of monitored network properties. Intuitively, we look to capture notions of: (i) *cut impact* which measures the expected load shed from the next failure; (ii) *transmission width* which quantify the difficulty of isolating a load from its nearest generators; (iii) *criticality* which measures the ability of grid components to adapt to future failures; and (iv) *load density* which reports the amount of load in a region of the network. The constructed attacks are provided to the **Interdiction Model** module, which plays the sequential interdiction game that fails components in the network one after another. The results of our exploration consist of a collection of attack sequences; we next analyze these sequences to gain insight into which *attack prefixes* are likely to lead to large disruptions in our **Impact Reasoning** module. We provide our SEQUIN framework as an open-source toolkit on GitHub [4].

3 Modeling Sequential Failures in the Grid

Next, we formalize the problem of identifying worst-case disruptions under *sequential* failures of key grid components. We first describe characteristics of the network, and how sequential component failures affect the grid. Then, we present the problem using an N-k interdiction model. A notation table is provided in App. A.

¹For the purpose of this paper, we model the dynamics of the power grid as a sequence of steady state models.

Electric Grid Infrastructure. We model the electric grid as a graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, where nodes represent buses, and edges represent branches or power lines. Each bus node $i \in \mathcal{N}$ is characterized by its voltage phase angle θ_i , which measures the shift in voltage (relative to a reference bus). We use two subsets of buses, $\mathcal{N}^{gen}$ and $\mathcal{N}^{load}$ to denote buses i that are co-located with: a generator producing at most $p_i^{g,+}$ power, or a load consuming p_i^d power. Each branch $(i, j) \in \mathcal{E}$ is characterized by a quantity representing the inverse of its reactance $b_{i,j}$, and its thermal rating $t_{i,j}$; these represent the difficulty and limits on power to be carried in the grid.

Dynamics of the Electric Grid. During its operation, the electric grid transports power produced at generators through lines to meet load demands in the network. This requires setting the generator power outputs, and determining how much is carried by each line. We define this formally using: generator setpoints p_i^g denoting the power output of a generator at bus i ; and power flows $p_{i,j}$ indicating the power carried on branch (i, j) . As changes occur (e.g., failures), generators modify their power output, also referred to as *ramping up/down*, over a set period of time. We assume that ramping cannot occur instantaneously, unless the generator is shut off. We capture this using δ_i , a bound on the change in power output from a setpoint p_i^g . Power flows in the grid are arbitrarily directed, since electricity flows in only one direction at any given time. Let $\mathcal{E}^f(i)$ and $\mathcal{E}^t(i)$ denote the edges that lead from or to bus i .

To maintain system operation after the failure of components in the grid, load may need to be cut from the network, e.g., if generators cannot supply the power needed to satisfy load demand. This can occur in the network manually (when directed by a grid operator), or automatically (when protection systems activate). Let p_i^s denote the amount of *load shed* from bus i . In the timescale considered for the sequential attack, we assume that load cannot be recovered after it is shed. This emulates the typical grid operator’s response, i.e., to avoid intermittently providing power to consumers.

Physics of Power Flows. The amount of flow carried through the grid is governed by two laws of physics: (i) *Kirchhoff’s Law* enforces the balance of flow at each bus in the network (Eq. 1a); and (ii) *Ohm’s Law* relates flows on branches to the difference in voltage phase angles across each branch (Eq. 1b). We assume that DC linear approximations are applied in the grid network, as in [31, 35], which approximates AC power with DC power.

$$p_i^g - p_i^d + p_i^s = \sum_{(i,j) \in \mathcal{E}^f(i)} p_{i,j} - \sum_{(j,i) \in \mathcal{E}^t(i)} p_{j,i} \quad \forall i \in \mathcal{N} \quad (1a)$$

$$p_{i,j} = b_{i,j}(\theta_i - \theta_j) \quad \forall (i,j) \in \mathcal{E} \quad (1b)$$

Sequential Failure Model. We initially assume that the grid is operating at full capacity, i.e., the grid is able to deliver power to all load demands. Suppose that after some time, one component fails. The grid then must reroute power flows and ramp generators up/down to compensate for the lost capacity, without violating laws of physics or engineering constraints. If the degraded network cannot meet load demands, then loads must be cut, i.e., load shedding. For the sequential component failures considered for this paper, suppose that failures occur one after another, and are interleaved with a network response, i.e., adjusting generator setpoints and power flows, and shedding load if needed. We assume that these failures can occur before or after the grid reaches steady state.

Since we are interested in the impact of each additional failure, we analyze the evolution of the network dynamics in discrete time epochs. To this end, let rounds $r \in 1, \dots, R$ denote each epoch. Then, we define $\underline{x}^r$ as the sequence of components in the grid $\mathcal{G}$ that have failed before or at round r . We let the full sequence, i.e., $\underline{x}^R$, have length k , where k is given. For ease of study, we assume that the k components of $\underline{x}^R$ are disabled across k fixed time epochs (i.e., $R=k$) and that some network response occurs after each failure. After each round, let $p_i^{g,r}$, $p_{i,j}^r$, and $p_i^{s,r}$ denote the generator setpoints, power flows, and load shed in the network, respectively.

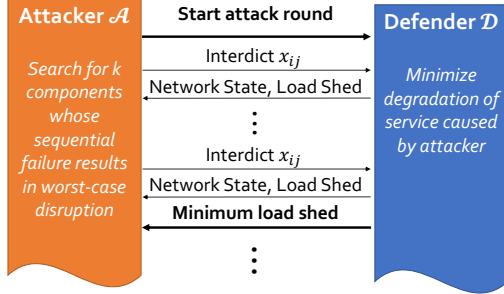


Figure 2: Sequential N-k interdiction model

Formally, the sequential N-k interdiction model is a tuple $(\mathcal{A}, \mathcal{D})$, where $\mathcal{A}$ is a sequential attacker, and $\mathcal{D}$ is the defender, as depicted in Fig. 2. To define these models correctly, care must be taken to prevent the defender from having a “lookahead” feature in which they know the attacker’s next actions. For a round r , the attacker’s and defender’s actions rely on the state of the network (defined using generator setpoints and branch power flows) in round $r-1$. This is due to how generators are bounded how much they can ramp up/down. Let σ^r denote the network state in round r , and $\sigma^r(\cdot)$ be the assignment of variable $(\cdot)$ in σ^r . We first define the defender model recursively, and then present the sequential attacker model.

Defender Model. The defender $\mathcal{D}$ is the player who aims to minimize the disruption to the grid. This is done by rerouting power flows and ramping generators up or down as part of the network response to a component failure. We assume that the defender’s actions are determined by a *max loadability* strategy, which aims to satisfy as much load demand in the network as possible. Then, we formulate the model for the defender in (2) as follows:

$$\mathcal{D}(\underline{x}; \sigma) = \min \sum_{i \in \mathcal{N}} (p_i^d - p_i^s) \quad (2a)$$

$$s.t. \quad p_i^g - p_i^d + p_i^s = \sum_{(i,j) \in \mathcal{E}^f(i)} p_{i,j} - \sum_{(j,i) \in \mathcal{E}^t(i)} p_{j,i} \quad \forall i \in \mathcal{N} \quad (2b)$$

$$p_{i,j} = b_{i,j}(\theta_i - \theta_j)x_{i,j}^r \quad \forall (i,j) \in \mathcal{E} \quad (2c)$$

$$p_i^g \leq \min\{\sigma^{r-1}(p_i^g) + \delta_i, p_i^{g,+}\} z_i \quad \forall i \in \mathcal{N}^{gen} \quad (2d)$$

$$p_i^g \geq \max\{0, \sigma^{r-1}(p_i^g) - \delta_i\} \quad \forall i \in \mathcal{N}^{gen} \quad (2e)$$

$$z_i \leq \sigma^{r-1}(z_i) \quad \forall i \in \mathcal{N}^{gen} \quad (2f)$$

$$-t_{i,j}(1 - x_{i,j}^r) \leq p_{i,j} \leq t_{i,j}(1 - x_{i,j}^r) \quad \forall (i,j) \in \mathcal{E} \quad (2g)$$

$$\sigma^{r-1}(p_i^s) \leq p_i^s \leq p_i^d \quad \forall i \in \mathcal{N}^{load} \quad (2h)$$

The defender’s goal is to minimize the total load shed across the grid (2a). We use (2b) and (2c) to represent the physics of power

flows. In round r , a generator at setpoint $\sigma(p_i^g)$ can ramp up/down by at most δ_i , but cannot fall lower than 0 or be greater than its maximum power output $p_i^{g,+}$. This is captured in (2e) and (2d). We use (2f) to express that after a generator shuts off, it cannot turn back on for the timescale of the sequential attack. The thermal rating of branches must also not be violated, as shown in (2g). We model that previously shed load cannot be recovered in (2h).

Attacker Model. The attacker $\mathcal{A}$ is a player whose goal is to cause the most disruption (i.e., load shed) to the grid. To this end, the attacker iteratively selects several sequences of k components in the grid to fail, and learns which sequences lead to larger load shed. In general, the attacker is not bounded by the number of sequences they can explore; however, complete enumeration is often computationally intractable (even with small k). Thus, the strategy of the attacker in efficiently selecting sequences of components to fail is crucial; we describe this as part of our approach later in Sec.4. To focus this paper, we assume that the attacker can only select transmission lines to interdict; extensions for other types of components can be made as in [31]. Thus, our binary interdiction variable will be index per branch (i,j) : $x_{i,j}^r = 1$ if (i,j) is interdicted by round r . We define the model of the attacker in (3) as follows:

$$\mathcal{A}(R; k) = \max_{r \in 1 \dots R} \sum_{i \in \mathcal{N}^{load}} \sigma^R(p_i^s) \quad (3a)$$

$$s.t. \quad \sum_{r \in 1 \dots R} x_{i,j}^r \leq 1 \quad \forall (i,j) \in \mathcal{E} \quad (3b)$$

$$\sum_{(i,j) \in \mathcal{E}} x_{i,j}^r = k^r \quad \forall r \in 1, \dots, R \quad (3c)$$

$$\underline{x}^r = \sum_{r' \leq r} \underline{x}_{i,j}^{r'} \quad \forall \underline{x}_{i,j}^{r-1} \in \underline{x}^{r-1} \quad (3d)$$

$$\sigma^r = \mathcal{D}(\underline{x}^r; \sigma^{r-1}) \quad \forall r \in 2, \dots, R \quad (3e)$$

The attacker’s objective in (3a) is to find the sequence $\underline{x}^r$ that maximizes the defender’s load shed at round R . This is constrained by cutting a line at most once, and the number of lines to cut per round, k^r , as expressed in (3b) and (3c). Note that the attacker is only able to add to the lines that are cut, i.e., once a line is failed, the decision cannot be reverted (3d). The defender’s actions across each round are captured by the attacker in (3e).

4 The SEQUIN Solution

This section presents the SEQUIN solution, which details: (i) a *Guided Exploration* module where we propose a strategy for the attacker to generate sequences of failures to examine; and (ii) an *Impact Reasoning* module where we analyze the load sheds from the explored sequences to gain insight into grid vulnerabilities. These modules are provided as a toolkit published on GitHub [4].

4.1 Guided Exploration

Our goal here is to inform the attacker’s strategy in the sequential N-k interdiction model. To this end, we first define several network properties which help indicate when the system may experience large disruptions. We then construct a set of attack sequences based on monitoring the state of the network for these properties. App. B provides algorithms for computing these network properties.

Definition: Cut Impact. The immediate potential to cause load shed is important in understanding which lines, when cut, produce the most disruption. We define the *cut impact* in the network to be the total load that a defender sheds due to a failure. This represents

a one round lookahead for the defender's response, as shown in Eq. 4. In general, a higher cut impact will yields more load shed.

$$CI(\mathcal{G}, \underline{x}^r, \sigma) = \max_{x_{i,j} \in \mathcal{E}} \mathcal{D}(\underline{x}^r \cup \{x_{i,j}\}; \sigma) \quad (4)$$

Definition: Criticality. Another key property of the state of the electric grid is its ability to re-route power through different paths in the event of failures. We capture this notion of flexibility by measuring the *criticality* of the generators and branches in the network. This is defined in Eq. 5a as the sum of the criticality for generators (Eq. 5b) and branches (Eq. 5c); a higher criticality indicates less flexibility in re-routing power to satisfy load demands.

$$C(\mathcal{G}, \sigma) = \sum_{i \in \mathcal{N}} \mathcal{G}C(i; \sigma) + \sum_{(i,j) \in \mathcal{E}} \mathcal{B}C(i, j; \sigma) \quad (5a)$$

$$\text{where } \mathcal{G}C(i; \sigma) = \min \left(|p_i^{g,-} - \sigma(p_i^g)|, |p_i^{g,+} - \sigma(p_i^g)| \right) \quad (5b)$$

$$\mathcal{B}C(i, j; \sigma) = |t_{i,j} - \sigma(p_{i,j})| \quad (5c)$$

Definition: Transmission Width. Load demands in the network that become isolated from generators nearest to them will typically need to be shed by the grid operator. Using this intuition, we derive the *transmission width* property, which aims to capture the number of ways that power can be supplied to a particular load. Eq. 6 models this as the minimum number of lines that need to be cut to isolate the node from its nearest generator. A smaller transmission width represents a load that can be disconnected easier.

$$\mathcal{T}W(i) = \min_{S \subseteq \mathcal{E}} |\{S : \forall g \in \mathcal{N} : \nexists \text{path}(g, i, \mathcal{G}-S)\}| \quad (6)$$

Definition: Load Density. Regions of the network where large loads exist are more likely to result in more load shed when attacked. We characterize the network in this manner using the *load density*, which is a measure of the total load in a topologically connected component of the network. Eq. 7 presents the load density, which is the sum of loads in the node's cluster. To construct each cluster, we group adjacent nodes with the same transmission width.

$$\mathcal{L}D(i) = \sum_{j \in \text{cluster}(i)} p_j^d \quad (7)$$

Generating Attack Sequences. Given these network properties, we next generate varying attack sequences to inform the attacker. Alg. 1 details this process. First, we find the initial steady state of the grid (line 1) and compute the previously mentioned network properties (lines 2-7). We first find the power flows and criticality of branches through the network (lines 3-5). Then, we note that the transmission width and the load density are used together, since part of the definition of load density includes grouping nodes in the network based on their transmission width. We compute the ratio of these two quantities as the value of the edges in the cluster (line 6-8). Intuitively, clusters with higher load density and lower transmission width are more impactful in the network and easier to isolate. In general, the sequences of attacks that are worth exploring are ones that contribute greatly to the operation of the network, e.g., high criticality. Thus, we take the top *expl* values from each of the computed sets to play the sequential N-k interdiction (lines 9-11). For each sequential failure, we consider using a truncated variant of the cut impact, which only contains lines whose power flows, criticalities, or ratio of transmission width to load density are high (line 14-22). This process is construct attack

Algorithm 1: Guided Exploration

Input: Graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, int k , int *expl*
Output: Attack Sequences $\mathcal{S}$
 // Put system into equilibrium
 1 $\sigma \leftarrow \text{SolveSteadyState}(\mathcal{G})$
 // Get network properties
 2 $pfs, crits, twld \leftarrow \text{map}(), \text{map}(), \text{map}()$
 3 **for** $(i, j) \leftarrow \mathcal{E}$ **do**
 4 $crits[(i, j)] \leftarrow \mathcal{B}C(i, j; \sigma)$
 5 $pfs[(i, j)] \leftarrow \sigma(p_{i,j})$
 6 **for** $i \in \mathcal{N}$ **do**
 7 $twld[i] \leftarrow \mathcal{L}D(i) / \mathcal{T}W(i)$
 8 $twld \leftarrow \text{MakeGroups}(twld)$
 // Extract 'expl' components from each
 9 $top_pfs \leftarrow \text{GetTopValues}(pfs, n = \text{expl})$
 10 $top_crits \leftarrow \text{GetTopValues}(crits, n = \text{expl})$
 11 $top_twld \leftarrow \text{GetTopValues}(twld, n = \text{expl})$
 // Generate attack sequences
 12 $\mathcal{S} \leftarrow \emptyset$
 13 $top_values \leftarrow \{top_pfs \cup top_crits \cup top_twld\}$
 14 **for** $x_0 \leftarrow top_values$ **do**
 15 $\underline{x} \leftarrow []$
 16 $nextcomps \leftarrow \emptyset$
 17 **for** $x \leftarrow 1, \dots, k$ **do**
 // Get next component to attack
 18 **if** $x == 1$ **then**
 19 $s.append(x_0)$
 20 **else**
 // Find component with largest cut impact from 'nextcomps'
 21 $x' \leftarrow CI((\mathcal{N}, \mathcal{E} - \underline{x}), nextcomps; \sigma)$
 22 $s.append(x')$
 // Evaluate interdiction model and update 'nextcomps' based on next state
 23 $\sigma' \leftarrow \text{PlayInterdictionModel}(\mathcal{G}, \underline{x}, \sigma)$
 24 $nextcomps \leftarrow \text{UpdateTopValues}(\sigma', top_values)$
 25 $\sigma \leftarrow \sigma'$
 26 $\mathcal{S} \leftarrow \mathcal{S} \cup \{\underline{x}\}$
 27 **return** $\mathcal{S}$

sequences iteratively, and selects the next component to fail based on the previous states of the network. The set of attack sequences is then collected and returned as output (lines 25-26).

4.2 Impact Reasoning

The explored attack sequences can then be analyzed to give insight into the different levels of load shed disruptions caused by sequential attacks. In particular, by aggregating sequences of failures based on their common prefixes, we can identify when and where component become crucial in the grid.

Analyzing Attack Sequences. To understand the range of impacts of the attacks explored in Alg. 1, we construct an attack tree that integrates prefixes of sequences with their resulting load shed disruption. In this attack tree, nodes represent the failure of a component and the resulting state of the grid, while edges represent

Algorithm 2: Analyze Attack Sequences

Input: Set of Attack Sequences $\mathcal{S}$, int k , Initial State σ
Output: Attack Tree $\mathcal{T}$
 // Initialize attack tree

```

1  $\mathcal{T} \leftarrow \{\sigma\}$ 
2 for  $l \leftarrow 1, \dots, k$  do
3   for  $s \leftarrow \mathcal{S}$  do
4     if  $l == 1$  then  $pa\_x \leftarrow \sigma$  else  $pa\_x \leftarrow s[l-1]$ ;
5      $x \leftarrow s[l]$ 
6     // Found common attack prefix
7     if  $(pa\_x, x) \in \mathcal{T}.getEdges(level = l)$  then
8        $prevsheds \leftarrow \mathcal{T}.get((pa\_x, x), l)$ 
9        $lo \leftarrow \min(s.loadshed, prevsheds.lo)$ 
10       $hi \leftarrow \max(s.loadshed, prevsheds.hi)$ 
11       $loadsheds \leftarrow (lo=lo, hi=hi)$ 
12    else
13      // Start new attack prefix
14       $loadsheds \leftarrow (lo=s.loadshed, hi=s.loadshed)$ 
15       $addEdge(\mathcal{T}, pa\_x, x)$ 
16       $addWeight(\mathcal{T}, pa\_x, loadsheds)$ 
17 return  $\mathcal{T}$ 
```

a transition to the next failed component. In this way, any path from the root of the tree to any node represents a prefix of a sequential attack. We annotate each node with a range of the potential future load shed disruption that can occur by further exploring paths that contain the node. Alg. 2 provides details on the construction of this tree, and how it can be used by a grid operator.

Our analysis occurs in two steps. First, the attack tree is constructed starting with a root node that assumes no failures have occurred (line 1). We then building the different levels of the attack tree based on the explored attack sequences. Each level l of the attack tree consists of nodes that represent the l th component failure in an attack sequence. The first level is constructed by adding an edge from the root node to new nodes representing the first element (i.e., component failure) in each sequence. Subsequent levels then add edges from level $l-1$ to l (lines 4-5). During the construction of this tree, we maintain a record of the minimum and maximum load shed disruption for each node (lines 6-12). We comment that thresholds for the load shed disruption can then be placed within the tree. By comparing the range of possible load shed values at one node, grid operators can quickly gain insight into how future failures impact the grid. Note that the attack tree is only run on the set of attack sequences found in the *Guided Exploration* phase of SEQUIN; thus, the attack tree will consider $|\mathcal{S}|$ attack sequences. We envision that this can be used to allow grid operators to potentially take action to help avoid or mitigate large scale disruptions.

The SEQUIN Toolkit. To help grid operators and stakeholders with studying potential sequential attacks on the grid, we also developed the SEQUIN toolkit, which allows visualization and exploration of different attack sequences and their respective impacts. Detailed instructions to install and use the SEQUIN tool are published on GitHub [4]. Fig. 3 depicts a grid with a few failed components. We envision that such a tool can be utilized to capture different attacks on the grid and the contribution that each component has towards the operation of the network.

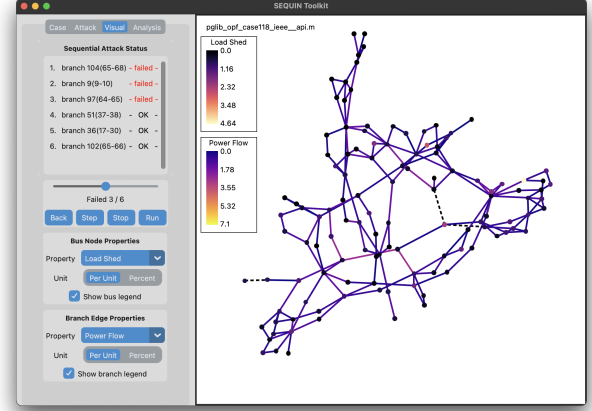


Figure 3: SEQUIN Toolkit

5 Experiments

In this section, we evaluate the SEQUIN approach on several benchmark power grid systems. Our experiments establish that varying the orderings of attacks can make a difference in the amount of load shed disruption in the grid. We also show the efficacy of the SEQUIN approach in generating attack sequences efficiently relative to other baseline comparisons.

Power Grid Datasets. We explore six benchmark power grid case studies of varying scale from the PGLib-OPF v23.07 library [39]. Three smaller networks with <100 nodes/edges and three larger-sized networks with >100 nodes/edges are utilized. Each network is normalized to present power in the “per-unit” base. Table 1 presents basic statistics about each network evaluated.

Table 1: Electric Grid Networks

Case Study	# Bus	# Branch	# Generators	Total Load (per-unit)
Case 14 IEEE API	14	20	5	4.63
Case 39 EPRI API	39	46	10	100.93
Case 60 C	60	88	23	89.40
Case 118 IEEE API	118	186	54	68.75
Case 162 IEEE DTC	162	284	12	72.39
Case 240 PSERC	240	448	143	1441.797

Experimental Setup. To simulate variation in load demands in each evaluated network, we modify the load p_i^d at each bus by randomly sampling a uniform distribution $\mathcal{U}(p_i^d * 95\%, p_i^d * 105\%)$, i.e., the load is modified to between 95% and 105% of its original value. Then, the network’s state (consisting of generator setpoints and branch power flows) is placed into equilibrium by solving the optimal power flow for the system. Our experiments use this equilibrium point as the initial state of the network. We conduct our experiments on an M1 MacBook Pro with 16 GB of memory and 10 CPU cores. The code was implemented in Julia using the JuMP [21] interface and the PowerModels [7] optimization library. Gurobi [17] is used for optimization, using a 1% optimality tolerance. Each result obtained reports the average across three runs. We publish our code implementation on GitHub [4].

Baseline Comparisons. We compared SEQUIN against several baseline techniques used for identifying failures. The standard baseline (STD) [36] assumes that lines are interdicted simultaneously (or near-simultaneously) without an intermediate network response. We note that this approach identifies the worst simultaneous failure under settings studied in most N-k interdiction models. The *permutation* baseline (PERM) iterates through all orderings of the solution from STD, and returns the one with the most load shed - a natural heuristic extension of simultaneous N-k. We also consider three greedy heuristic baselines for generating attack sequences: (i) *Greedy-FLOW* which selects lines based on its current power flow; (ii) *Greedy-SHED* which selects lines based on the resulting load shed; and (iii) *Greedy-CRIT* which selects lines to increase criticality. To allow each of the greedy baselines to explore more solutions, the top 10% of lines that maximize the greedy heuristic are also tested, e.g., the top 10% of lines with highest power flows would be evaluated for *Greedy-FLOW*. Finally, for small networks and/or small k , we additionally report the *enumeration* baseline (ENUM), which exhaustively explores all possible orderings of attacks, and returns the one with the most load shed. This effectively captures the worst-case disruption for any setting, but becomes computationally intractable as the network scale grows. As described earlier, all disruption to the grid is measured using load shed.

5.1 Impact of Sequential Attacks

Our first experiment examines the impact of considering a sequential attack on the grid, in comparison with the standard attack where all selected components fail simultaneously. To this end, we compare the load shed from the STD solution to the distribution of load sheds from orderings of the PERM solution, and when possible, the ENUM solution. Note that the PERM solution interdicts the same lines as the STD solution, but just introduces an order to do so. Fig. 4 depicts histograms of these load sheds for two illustrative networks. For the small network, we set $k = 3$ and a generator ramping bound of $\delta = 10\%$. The larger network is evaluated with $k = 6$ and $\delta = 1\%$. We note that a smaller value of k is chosen for the small network to include the ENUM solution baseline.

Our results shows how sequential failures can potentially cause near or more disruption than their simultaneous counterparts. In Fig. 4a, we plot the histogram of the ENUM solution for the small network (blue bars), and mark the locations of the STD solution (vertical red line) and the highest PERM solution (dotted vertical blue line). While the STD solution lies 7.5% above the highest PERM solution, the highest ENUM solution lies 19.3% above the STD solution. This shows that care must be taken in selecting sequences of failures, as a majority of the sequences from the ENUM solution yield little to no load shed disruption. The network in Fig. 4b also reports a similar result. Here, we plot the distribution of the PERM solution (blue bars), and the location of the corresponding STD solution (vertical red line); the sequences from the ENUM solution were too expensive to compute for this case study. Notably, we find that 81.4% of these permutation solutions lie at or above the STD solution, and the highest of these values is 47% larger.

Note that the methods are not completely comparable; each failure model assumes a different defender model. That is, the aggregate network response between the simultaneous and sequential

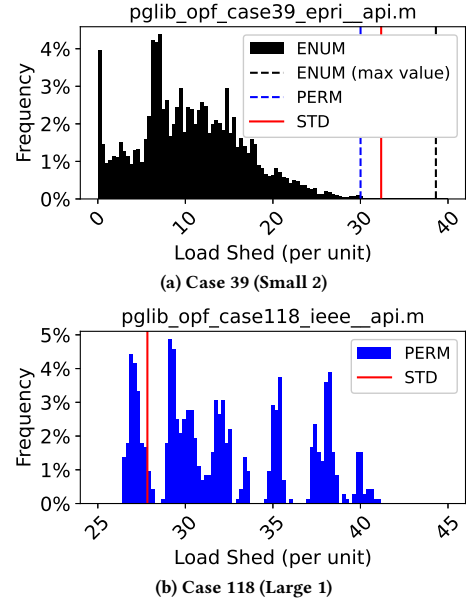


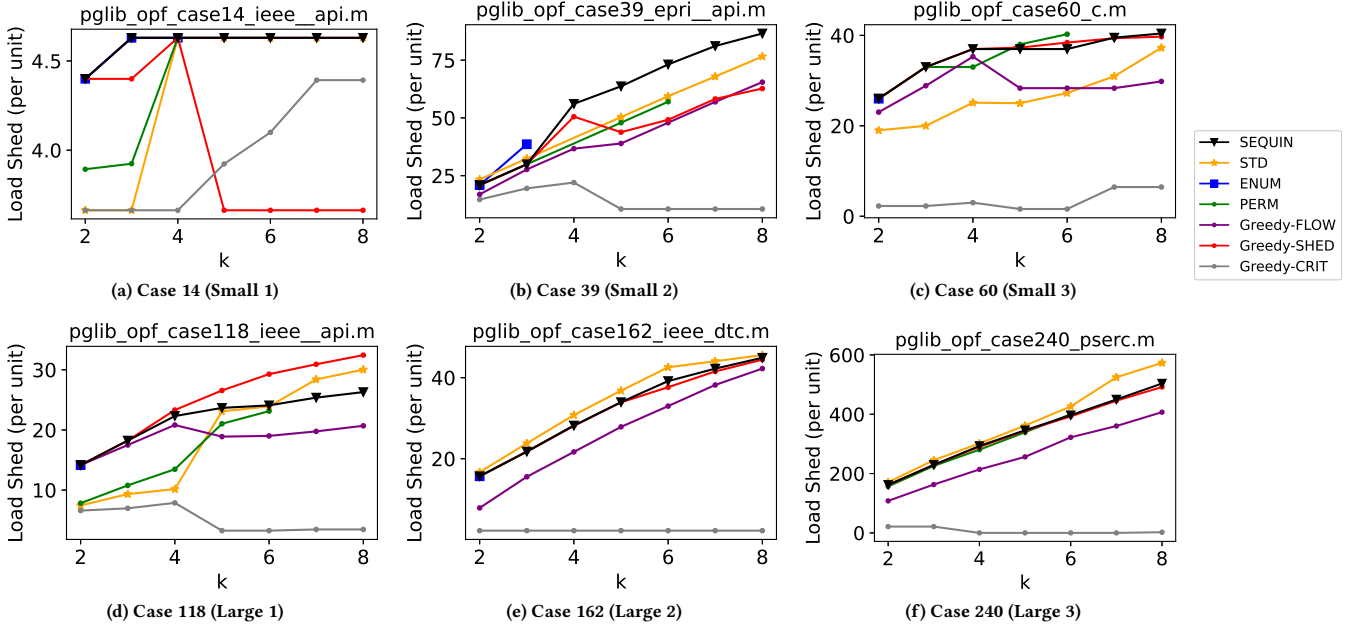
Figure 4: Distribution of Load Shed (per-unit) between Sequential and Simultaneous Attack

failure models allow for different network states to be reached, i.e., a generator in the simultaneous failure model will ramp up or down by at most δ , while the sequential model allows generators to ramp up/down by $(k - 1) * \delta$. Despite this, it is still useful to look at these differences, as it reveals that the sequence of an attack can vary greatly from its simultaneous counterpart. Our results show that there can be sequences of failures that produce larger disruptions, motivating the need to study and understand such failures.

5.2 Varying the Number of Interdicted Components

We next explore how changing the total number of interdicted components, k , affects the maximum load shed in round R . In each network, we fix the generator ramping bound to $\delta = 10\%$ for all generators. Fig. 5 plots the maximum load shed found across SEQUIN and the other baseline strategies, from $k = 2$ to $k = 8$. Larger values of k indicate longer sequences of failures in the grid.

In general, we observe that as k increases, the average load shed across all networks also increases (as long as the interdicted components are chosen correctly). The average percent change between the sequences found by SEQUIN, and the sequential baseline methodologies (excluding *Greedy-CRIT*) range from 0.90 – 38.85% for the small networks, and 0.12 – 25% for the large networks. We exclude *Greedy-CRIT* from these percent changes computed, since it generally seems to perform very poorly, and skews results. This shows that our approach is consistently able to produce impactful attack sequences to study. We notice that the lower bound on this percent change is small, which is due to how the some greedy heuristics used yield similar sequences to SEQUIN. However, this does not necessarily stay true across all different values of k evaluated. We will also show later that SEQUIN is able to trade off the time taken to compute these results in Sec.5.4.

Figure 5: Number of lines interdicted (k) vs Load Shed (per-unit)

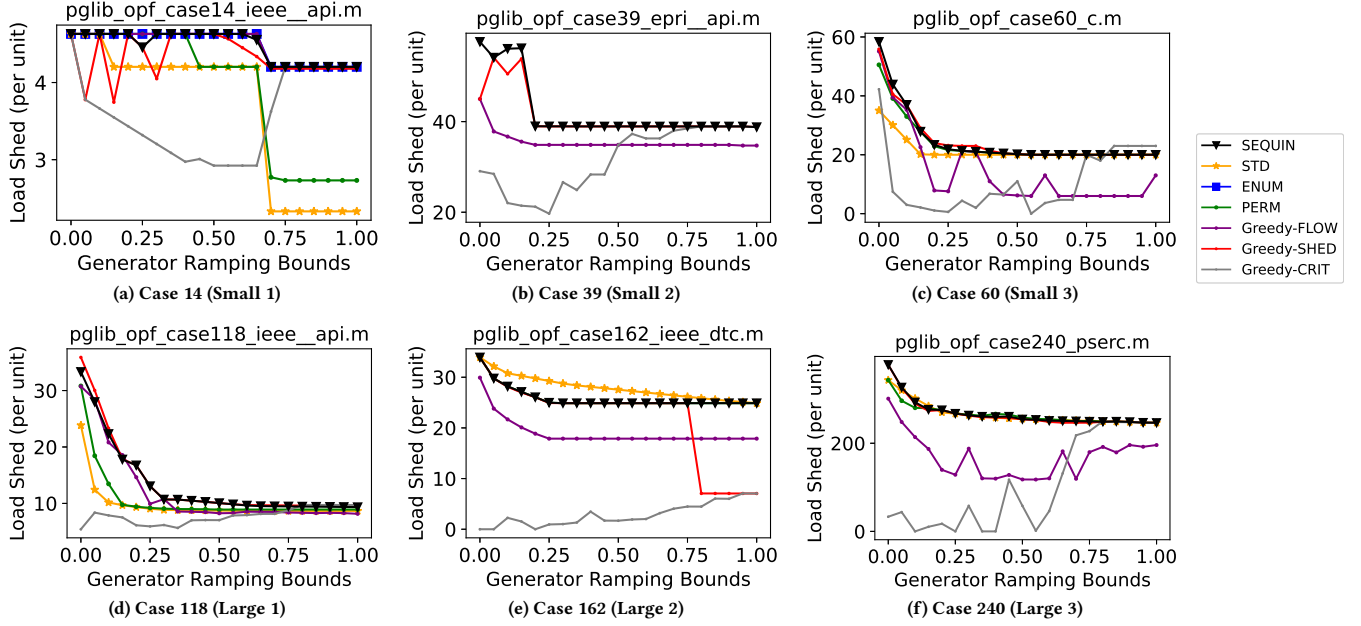
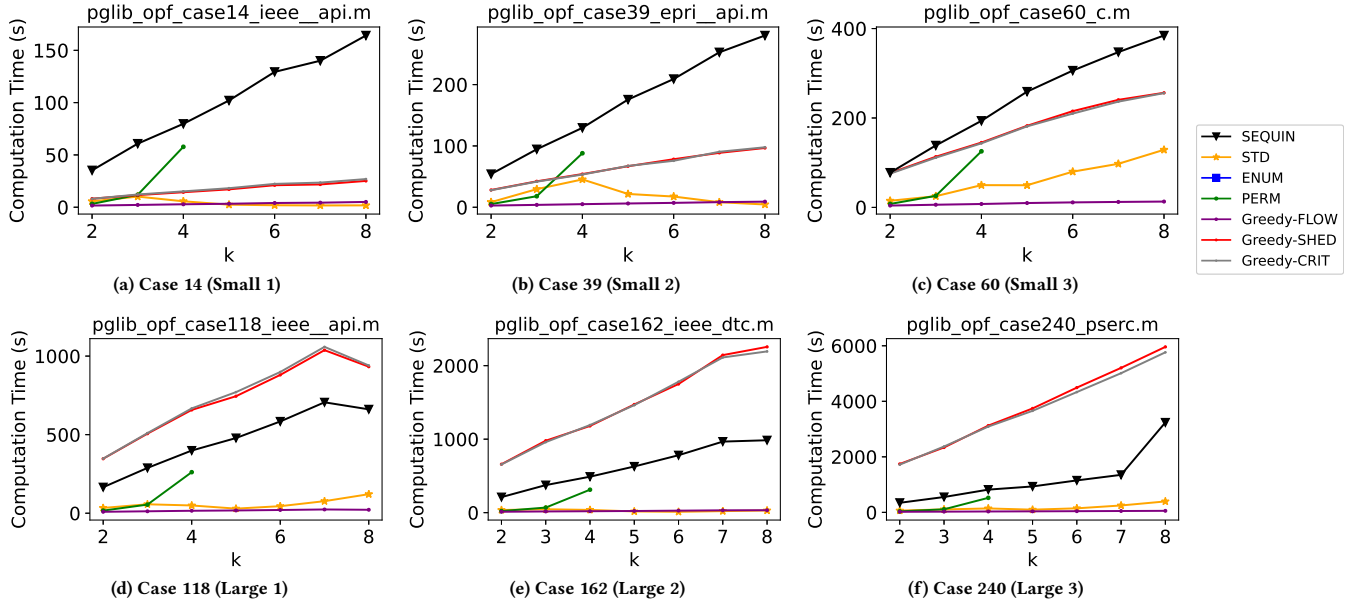
Examining some of these networks in turn for the highest load shedding sequence found, we notice first that for very small networks, particularly Case 14, the load shed increase only occurs until $k = 3$. This behavior is expected when the entire grid fails due to the interdicted lines. Thus, as k becomes larger, the load shed in the network does not grow. This is observed using the ENUM baseline. For the other two small networks, we observe one of two patterns emerge. In Case 39, the general load shed increases (close to) linear as k increases, while in Case 60, the rate of increase slows down after $k = 4$. Intuitively, these patterns depict how either additional failures in the grid can still be impactful; or have marginal load shed in the network. While this indicates that certain k may be more useful to study for some networks, i.e., the problem of determining k as a resilience factor for the grid, this is out of scope for this paper. Larger networks also tend to follow this pattern of linear increase.

5.3 Varying Generator Ramping Bounds

We also explore the impact of varying the generator ramping bounds on the total load shed at round R for each network. The generator's ramping bounds are a proxy measure of the elapsed time in between the failure of components in the grid: a smaller ramping bound corresponds to a shorter time period. Fig. 6 shows plots of the amount of load shed found by the SEQUIN approach and each baseline method when δ is varied from 0% (where the generator cannot ramp up or down), to 100% (where the generator can ramp to any output below its maximum $p^{g,+}$). In this experiment, we consider the ramping bound at a step size of 5% (i.e., we examine load shed at every 5%), and set $k = 4$ for each result.

In regards to the resulting load shed values found, we note that our approach is able to consistently identify sequences of failures that yield in high load shed. This load shed is similar to that of the greedy load shedding heuristic used; however, this is done in significantly less time, as shown later in Sec.5.4.

This experiment reveals several interesting patterns that depict different behaviors of the grid depending on the value of the generator ramping bounds. For ease of explanation, we use Case 118 in Fig. 6, as this illustrates the general pattern best. At $\delta = 100\%$, we find that all solutions, regardless of whether there is a simultaneous or sequential failure, will produce the same load shed. This is because generators are allowed to ramp up or down to their optimal/natural setpoints at each round. From this perspective, the last round R in a sequence of failures becomes effectively indistinguishable from the simultaneous case. This notion of the *natural setpoint*, which is the value of a generator's power output, describes the behavior of the grid from between $\delta = 100\%$ to $\delta \approx 25\text{--}50\%$. In effect, this range of ramping bounds largely does not constrain the ability of the generator to ramp up or down, and so there is little to no divergence between the simultaneous and sequential failures. The next set of behaviors in the grid occur between $\delta \approx 25\text{--}50\%$ (at the natural setpoints of generators), and $\approx 10\%$ (where the next set of behaviors of the grid begins). Notably, the amount of load shed disruption across the simultaneous and sequential failures starts to split. Here, we observe that the PERM solution generally is generally lower than the STD solution; as hinted earlier, this is due to the orderings considered by PERM being suboptimal to cut for the now evolving network. This is evidenced by the other strategies that are able to perform better than PERM. However, we note that the sequences explored by the greedy approaches are only simple heuristics, and thus sometimes jump in the load shed found (as observed in Fig. 6). Finally, at $\delta = 0 - 10\%$, we find that solutions begin to converge again, albeit at a much larger load shed value. At the extreme ($\delta = 0\%$), generators are unable to ramp at all, which yields higher load shed disruption. Also note that with stricter bounds, when a generator is unable to ramp up/down fast enough, it is shut down; this then requires more load to be cut.


 Figure 6: Number of lines interdicted (k) vs Load Shed (per-unit), $k=4$

 Figure 7: Number of lines interdicted (k) vs Computation Time (s)

5.4 On Computational Runtime

We lastly report on the average time taken to find the worst-case disruption events for the SEQUIN approach, and compare it to other baseline techniques. Fig. 7 shows the time taken for $k = 2$ to $k = 8$, for each benchmark network (aggregated across all generator bounds). The ENUM strategy is excluded since many of the cases are unable to fully enumerate across all solutions.

As expected, it generally takes longer to search for impactful sequences when k grows, or when larger networks are evaluated. A key observation is that our approach will generally do more work

to find potential sequences, which causes it to take longer for the smaller sized networks. However, the process taken by SEQUIN pays off for the larger sized networks, as our approach is able to adequately explore different sequences in the network. We attribute this primarily to the network science properties that are used to indicate which parts of the network would likely cause significant network state shift. Comparing results between the smaller and larger sized networks, note that small networks typically all have run times that fall under 400s, while the larger networks tend to scale more poorly. To avoid extreme outliers, we limit the PERM

baseline to $k = 4$, since it scales exponentially in the size of k . We find that the Greedy-FLOW baseline performs similarly across different values of k ; this is because it only evaluates the state of the network once before selecting its top components for interdiction. Between our approach and the other baselines, note that there is significantly more work done in examining the state of the network - and thus can sequences can be better identified. Finally, we observe that SEQUIN seems to consider the tradeoffs between exhaustively searching for attack sequences and the time taken for searching, as evidenced by the lower run times, combined with the larger load sheds found as discussed previously.

6 Conclusions and Future Work

This paper presented SEQUIN, a framework that integrates network science and physics-based optimization into a SEQUential N-k Interdiction model to solve the problem of identifying worst-case sequential disruptions to electric grid infrastructures. Our approach efficiently informs the attacker using an interdiction model of which components would lead to large load shed disruptions. We provide a methodology to aggregate and reason about the impacts of different types of attacks in the grid. We envision that this can then be utilized by grid operators to understand worst-case outcomes. To aid in the usability of SEQUIN, we developed a open-source interactive toolkit that showcases these concepts. Future work will explore how such analysis and tools can be utilized by grid operators to ruggedize the grid and minimize the impact of sequential attacks. We also aim to improve the identification of such attacks under more dynamic constraints, e.g., by considering failures with variable time epochs, in larger use cases. More work in scaling exact decomposition methods for simultaneous N-k failure can also help in identifying vulnerabilities.

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References

- [1] H. Bai and S. Miao. 2015. Hybrid flow betweenness approach for identification of vulnerable line in power system. *IET Gener. Transm. Distrib.* 9, 12 (2015).
- [2] R.S. Biswas et al. 2020. A graph theoretic approach to power system vulnerability identification. *IEEE TPWRS* 36, 2 (2020), 923–935.
- [3] S.R. Chestnut and R. Zenklusen. 2017. Hardness and approximation for network flow interdiction. *Networks* 69, 4 (2017), 378–387.
- [4] A. Chio. 2025. SEQUIN. <https://github.com/andrewgchio/SEQUIN>
- [5] P. Chopade and M. Bikkash. 2016. New centrality measures for assessing smart grid vulnerabilities and predicting brownouts and blackouts. *IJCIP* 12 (2016).
- [6] P. Cicilio et al. 2020. Electrical grid resilience framework with uncertainty. *Electr. Power Syst. Res.* 189 (2020), 106801.
- [7] C. Coffrin et al. 2018. PowerModels.jl: An Open-Source Framework for Exploring Power Flow Formulations. In *PSCC*. 1–8.
- [8] H. Davarikia and M. Barati. 2018. A tri-level programming model for attack-resilient control of power grids. *J. MPCE* 6, 5 (2018), 918–929.
- [9] N.L. Dehghani et al. 2021. Intelligent hurricane resilience enhancement of power distribution systems via deep reinforcement learning. *Applied energy* 285 (2021).
- [10] A. Delgadillo, J.M. Arroyo, and N. Alguacil. 2009. Analysis of electric grid interdiction with line switching. *IEEE TPWRS* 25, 2 (2009), 633–641.
- [11] V. Donde et al. 2008. Severe multiple contingency screening in electric power systems. *IEEE TPWRS* 23, 2 (2008), 406–417.
- [12] D. Dwivedi et al. 2023. Evaluating the planning and operational resilience of electrical distribution systems with distributed energy resources using complex network theory. *Renewable Energy Focus* 46 (2023), 156–169.
- [13] R. Eskandarpour, A. Khodaei, and A. Arab. 2017. Improving power grid resilience through predictive outage estimation. In *NAPS*. IEEE, 1–5.
- [14] N. Fan et al. 2011. Economic analysis of the n-k power grid contingency selection and evaluation by graph algorithms and interdiction methods. *Energy Systems* 2, 3 (2011), 313–324.
- [15] J. Fang et al. 2016. Power system structural vulnerability assessment based on an improved maximum flow approach. *IEEE Trans. Smart Grid* 9, 2 (2016), 777–785.
- [16] J. Gao et al. 2015. Recent progress on the resilience of complex networks. *Energies* 8, 10 (2015), 12187–12210.
- [17] Gurobi Optimization, LLC. 2024. Gurobi Optimizer Reference Manual. <https://www.gurobi.com>
- [18] M.M. Hosseini and M. Parvania. 2021. Resilient operation of distribution grids using deep reinforcement learning. *IEEE Trans. Ind. Inform.* 18, 3 (2021).
- [19] N.U. Ibne Hossain et al. 2020. Modeling and assessing cyber resilience of smart grid using Bayesian network-based approach: a system of systems problem. *JCDE* 7, 3 (2020), 352–366.
- [20] D. H. Kim et al. 2017. Network topology and resilience analysis of South Korean power grid. *Phys. A: Stat. Mech. Appl.* 465 (2017), 13–24.
- [21] Miles L. et al. 2023. JuMP 1.0: Recent improvements to a modeling language for mathematical optimization. *MPC* (2023).
- [22] X. Liu et al. 2016. Microgrids for enhancing the power grid resilience in extreme conditions. *IEEE Trans. Smart Grid* 8, 2 (2016), 589–597.
- [23] X. Liu and Z. Li. 2014. Local load redistribution attacks in power systems with incomplete network information. *IEEE Trans. Smart Grid* 5, 4 (2014), 1665–1676.
- [24] J.M. López-Lezama et al. 2017. Assessment of the Electric Grid Interdiction Problem using a nonlinear modeling approach. *Electr. Power Syst. Res.* 144 (2017).
- [25] S. Maharjan et al. 2013. Dependable demand response management in the smart grid: A Stackelberg game approach. *IEEE Trans. Smart Grid* 4, 1 (2013), 120–132.
- [26] A. Mastin et al. 2023. Best Response Intersection: An Optimal Algorithm for Interdiction Defense. *IJOO* 5, 2 (2023), 172–190.
- [27] M. Panteli et al. 2017. Metrics and quantification of operational and infrastructure resilience in power systems. *IEEE TPWRS* 32, 6 (2017), 4732–4742.
- [28] Z. Qin, Q. Li, and M.C. Chuah. 2012. Unidentifiable attacks in electric power systems. In *ACM/IEEE ICCPS*. IEEE, 193–202.
- [29] R. Rocchetta. 2022. Enhancing the resilience of critical infrastructures: Statistical analysis of power grid spectral clustering and post-contingency vulnerability metrics. *Renew. Sustain. Energy Rev.* 159 (2022), 112185.
- [30] V. Rosato, S. Bologna, and F. Tiriticco. 2007. Topological properties of high-voltage electrical transmission networks. *Electr. Power Syst. Res.* 77, 2 (2007).
- [31] J. Salmeron, K. Wood, and R. Baldick. 2009. Worst-case interdiction analysis of large-scale electric power grids. *IEEE TPWRS* 24, 1 (2009), 96–104.
- [32] L. Shen et al. 2020. Large-scale systems resilience: A survey and unifying framework. *QREI* 36, 4 (2020), 1386–1401.
- [33] J. Stiasny, G.S. Misyris, and S. Chatzivasileiadis. 2021. Transient stability analysis with physics-informed neural networks. *arXiv preprint arXiv:2106.13638* (2021).
- [34] K. Sundar et al. 2019. Spatial and topological interdiction for transmission systems. *arXiv preprint arXiv:1904.08330* (2019).
- [35] K. Sundar et al. 2021. Credible interdiction for transmission systems. *IEEE Trans. Control Netw Syst* 8, 2 (2021), 738–748.
- [36] K. Sundar et al. 2024. Exact and Heuristic Approaches for the Stochastic N-k Interdiction in Power Grids. In *PMAPS*. 1–6.
- [37] R. Tan et al. 2016. Optimal false data injection attack against automatic generation control in power grids. In *ACM/IEEE ICCPS*. IEEE, 1–10.
- [38] The Fayetteville Observer. 2022. 'Intentional, willful and malicious': 40K without power after substation attacks in Moore County.
- [39] The IEEE PES Task Force on Benchmarks for Validation of Emerging Power System Algorithms. [n.d.]. PGLib Optimal Power Flow Benchmarks. <https://github.com/power-grid-lib/pglib-opf>
- [40] H. Wang et al. 2019. Dynamic data injection attack detection of cyber physical power systems with uncertainties. *IEEE Trans. Ind. Inform.* 15, 10 (2019).
- [41] E.B. Watson et al. 2019. Modeling electrical grid resilience under hurricane wind conditions with increased solar and wind power generation. *TPWRS* 35, 2 (2019).
- [42] M. Xu et al. 2024. Resilient dynamic microgrid formation by deep reinforcement learning integrating physics-informed neural networks. *Eng. Appl. Artif. Intell.* 138 (2024), 109470.
- [43] J. Yan, Y. Tang, B. Tang, H. He, and Y. Sun. 2016. Power grid resilience against false data injection attacks. In *IEEE PESGM*. IEEE, 1–5.
- [44] N. Yodo, P. Wang, and Z. Zhou. 2017. Predictive resilience analysis of complex systems using dynamic Bayesian networks. *IEEE Trans. Reliab.* 66, 3 (2017).
- [45] M. Yu and S. H. Hong. 2015. A real-time demand-response algorithm for smart grids: A Stackelberg game approach. *IEEE Trans. Smart Grid* 7, 2 (2015), 879–888.
- [46] L. Zhao and B. Zeng. 2013. Vulnerability analysis of power grids with line switching. *IEEE TPWRS* 28, 3 (2013), 2727–2736.

A Tables of Notations

Table 2: Notation for the Electric Grid

Symbol	Definition
$\mathcal{G}$	Graph of the network
$i \in \mathcal{N}$	Set of buses (nodes)
$(i, j) \in \mathcal{E}$	Set of branches / lines (edges)
$\mathcal{N}^{gen}$	Set of buses co-located with a generator
$\mathcal{N}^{load}$	Set of buses co-located with a load demand
$\mathcal{E}^f(i)$	Set of edges directed from bus i
$\mathcal{E}^t(i)$	Set of edges directed to bus i
Variables	θ_i Voltage phase angle at bus i
	p_i^g Active power generated at bus i
	δ_i Ramping bounds for generator at bus i
	p_i^s Active power shed from bus i
	$p_{i,j}$ Active power flow on branch (i, j)
Constants	p_i^d The power consumed at i
	$p_i^{g,+}$ Maximum power generated at bus i
	$b_{i,j}$ Inverse of the reactance in branch (i, j)
	$t_{i,j}$ Thermal limit of branch (i, j)
Interdiction	$\mathcal{A}$ Attacker Player in the interdiction model
	$\mathcal{D}$ Defender Player in the interdiction model
	k The number of components interdicted
	r Time epoch between attacker's sequential failures
	R Maximum number of rounds
	$x_{i,j}^r$ Binary variable denoting if line (i, j) is interdicted
	$\underline{x}^r$ A vector of binary variables $\{x_{i,j}^r\}$

B Computation of Network Properties

This appendix presents algorithms that can be used to compute each of the network science properties introduced in Sec. 4.

Algorithm 3: Compute Cut Impact

Input: Graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, State σ
Output: Cut Impact $CI(\mathcal{G})$

```

1  $cmpt, impt \leftarrow \text{null}, 0$ 
2 for  $(i, j) \leftarrow \mathcal{E}$  do
3   if  $impt < \mathcal{D}(\underline{x} \cup \{x_{i,j}\}; \sigma)$  then
4      $cmpt, impt \leftarrow x_{i,j}, \mathcal{D}(\underline{x} \cup \{x_{i,j}\}; \sigma)$ 
5 return  $impt, cmpt$ 
    
```

Algorithm 4: Compute Criticality

Input: Graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, State σ
Output: Criticality $C(\mathcal{G})$

```

1  $gencrit \leftarrow \sum_{i \in \mathcal{N}^{gen}} \min(|p_i^{g,-} - \sigma(p_i^g)|, |p_i^{g,+} - \sigma(p_i^g)|)$ 
2  $brcrit \leftarrow \sum_{(i,j) \in \mathcal{E}} |t_{i,j} - \sigma(p_{i,j})|$ 
3 return  $gencrit + brcrit$ 
    
```

Algorithm 5: Compute Transmission Width

Input: Graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, Node i
Output: Transmission Width $\mathcal{TW}(i)$

```

1  $width, remove \leftarrow 0, \emptyset$ 
2 for  $gen \leftarrow \mathcal{N}^{gen}$  do
3   while True do
4     // Find a path from generator to node  $i$ 
5      $\mathcal{G}' \leftarrow (\mathcal{N}, \mathcal{E} - remove)$ 
6      $p \leftarrow \text{path}(gen, i, \mathcal{G}')$ 
7     if  $p == \text{null}$  then break;
8      $remove \leftarrow remove \cup \text{edges}(p)$ 
9      $width \leftarrow width + 1$ 
9 return  $width$ 
    
```

Algorithm 6: Compute Load Density

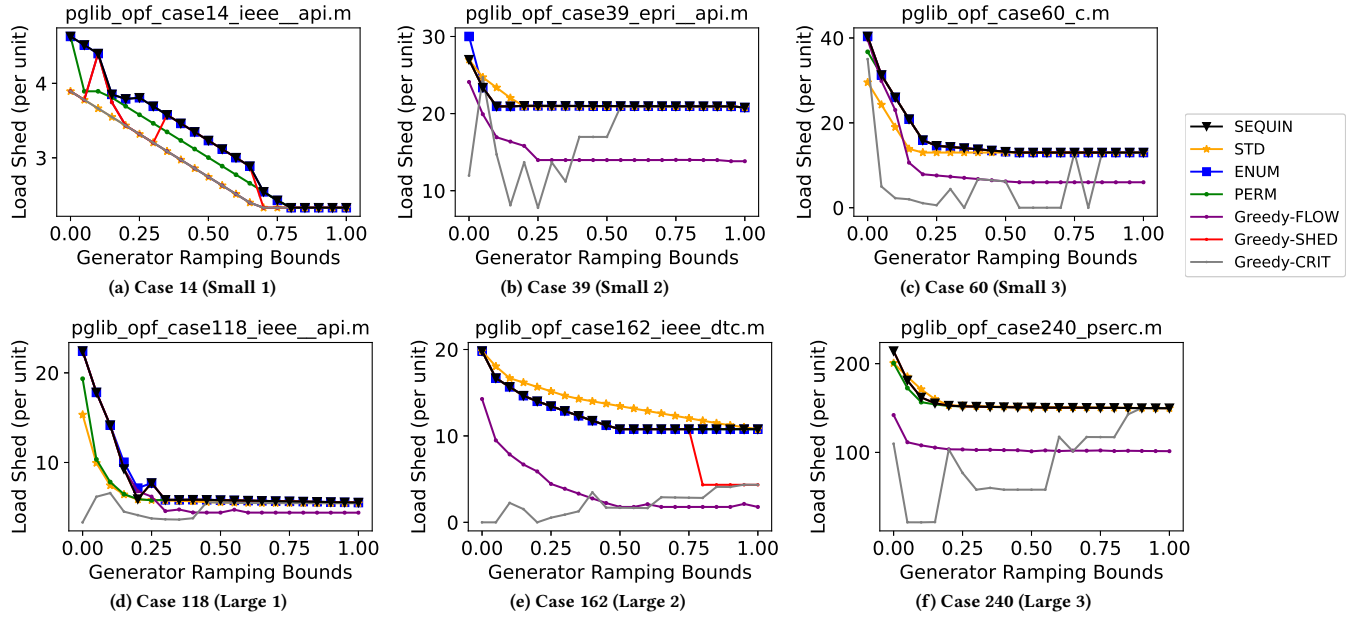
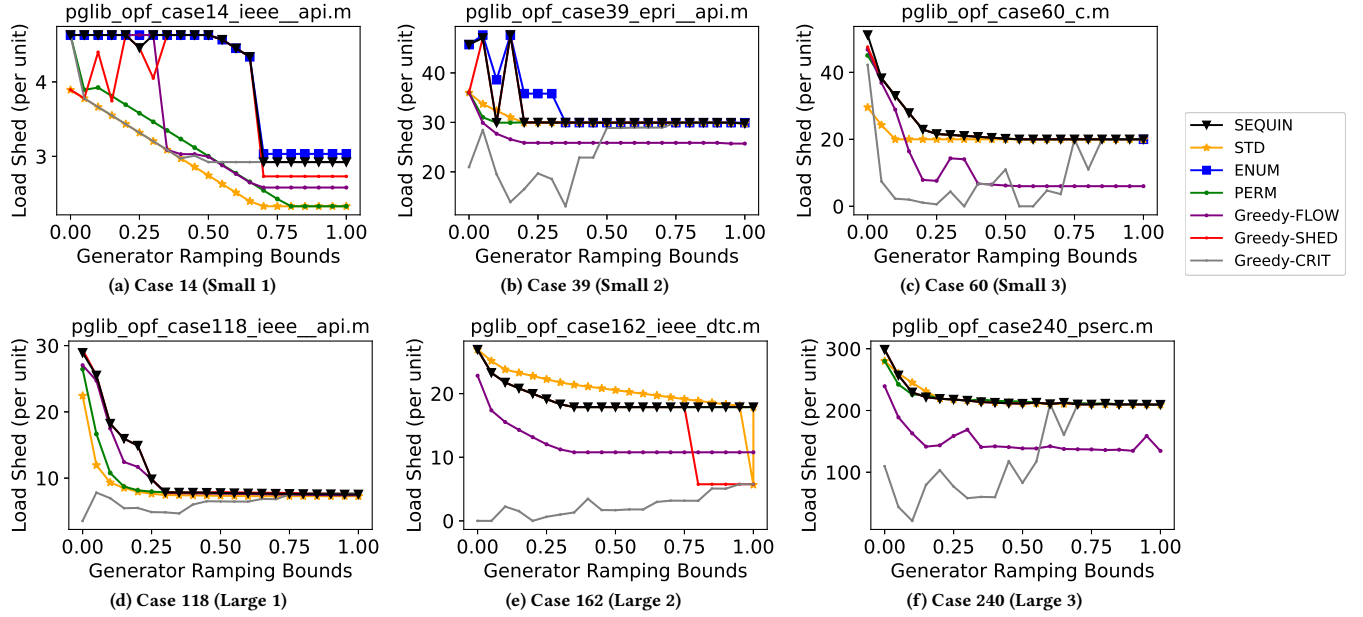
Input: Graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, Node i
Output: Load Density $\mathcal{LD}(i)$

```

1  $tw, clusters \leftarrow \text{map}(), \text{map}()$ 
  // Initialize Transmission Width and Clusters
2 for  $i \leftarrow \mathcal{N}$  do
3    $tw[i] \leftarrow \text{ComputeTransmissionWidth}(\mathcal{G}, i)$ 
4    $clusters[i] \leftarrow \{i\}$ 
  // Group by Similar Transmission Width
4 for  $(i, j) \leftarrow \mathcal{E}$  do
5   if  $tw[i] == tw[j]$  then
6      $\text{Merge}(clusters[i], clusters[j])$ 
7  $totalLoad \leftarrow \sum_{j \in clusters[i]} p_i^d$ 
8 return  $totalLoad$ 
    
```

C Additional Experiments for Varying Generator Ramping Bounds, with $k=2$ and $k=3$

We lastly present additional experiments that were run for varying the generator ramping bounds (similar to Sec. 5.3) in Fig. 8 and Fig. 9. These figures differ in the choice of $k = 2$ or $k = 3$. The general trends observed in Sec. 5.3 are still found in the figures.


 Figure 8: Number of lines interdicted (k) vs Load Shed (per-unit), $k=2$

 Figure 9: Number of lines interdicted (k) vs Load Shed (per-unit), $k=3$